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A Multihead Self-Attention and Adaptive Multiscale Dilated Convolution Model for Generalizable Arrhythmia Classification from ECG Signals

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Abstract

Electrocardiogram (ECG) signals exhibit distribution shifts across recording devices, populations, sampling rates, and lead configurations, impairing generalization and long-term feature extraction. To address these issues, this paper proposes a multihead self-attention and adaptive multiscale dilated convolution model (MHA-MSD). First, a multihead encoder projects ECG features into a high-dimensional space to encode both temporal dynamics and inter-lead correlations. A subsequent self-attention mechanism then enhances cross-dataset generalization. Next, an adaptive multiscale dilated convolution module constructs a multibranch network with different dilation rates and uses a gating network to dynamically adjust branch weights according to signal length and feature distribution. Finally, a spatial attention module fuses multibranch features and refines features during dimensionality reduction to enhance global feature learning. On CPSC2018, the model classifies nine categories (Normal, AF, I-AVB, LBBB, RBBB, PAC, PVC, STD, and STE), achieving 93.6% accuracy and a 92.9% mean F1 score in the best-performing fold among the 10 cross-validation iterations. On the MIT-BIH-to-INCART transfer, the model classifies normal (N), supraventricular ectopic (S), and ventricular ectopic (V) beats and achieves 97.38% accuracy and 97.41% mean F1 score, while class-discriminative attention maps align with clinically relevant waveform segments. Cross-dataset classification of N, LBBB, and RBBB among CPSC2018, PTB-XL, and CODE-15% yields respective F1 scores of 0.87, 0.96, and 0.86 for CPSC2018-to-PTB-XL transfer.

Keywords: computational modeling; data-driven methods; predictive analytics; computer-aided decision support; arrhythmia diagnosis; electrocardiography; deep learning

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1. Introduction

Over the past two decades, global arrhythmia incidence has remained persistently high. This imposes severe burdens on patients and families. Long-term, reliable arrhythmia monitoring is essential, as it can reduce the clinical burden of the disease and improve quality of life for at-risk individuals. Electrocardiogram (ECG) analysis is the primary method for the diagnosis of arrhythmias. A multi-lead ECG provides a comprehensive assessment of cardiovascular health.

Real-time expert analysis of massive multi-lead ECG data for arrhythmia detection and cardiovascular monitoring remains impractical. Developing automated arrhythmia diagnosis algorithms has become a critical research focus. Advancements in machine learning and deep learning have enabled researchers to achieve promising results in arrhythmia detection. For example, self-supervised learning that dynamically fuses time- and frequency-domain information has recently been explored for atrial fibrillation detection [1]. Recent architectural innovations in deep learning, particularly transformer-based networks, demonstrate unprecedented performance in arrhythmia classification [2]. However, substantial differences in ECG signal characteristics and recording conditions across patients and datasets can impair cross-dataset generalization [3,4]. In addition, methods centered on individual heartbeats or predefined signal segments [5,6] may not fully exploit temporal context beyond the selected analysis window. These challenges motivate architectures that jointly model temporal dependencies, inter-lead information, and multiscale ECG features.

To address these challenges, we propose a multihead self-attention and adaptive multiscale dilated convolution network, named MHA-MSD, to improve the generalization and accuracy on arrhythmia detection. The multihead self-attention module tackles cross-dataset generalization limitations, while the adaptive multiscale dilated convolution module resolves inadequate classification capabilities for long-range ECG signals. The main contributions of this work are as follows.

1. A multihead self-attention module is proposed to establish representations and relationships of temporal patterns and inter-lead characteristics through ECG signal vectorization in high-dimensional space. This module significantly enhances cross-dataset generalization capabilities by integrating time-domain features with inter-lead characteristics during model training.
2. An adaptive multiscale dilated convolution module, together with gated branch weighting and spatial attention fusion, integrates local multiscale morphological features with the global contextual representations learned by the self-attention front end.
3. Evaluation on CPSC2018, MIT-BIH [7], INCART [8], PTB-XL [9], and CODE-15% [10] assesses both classification performance and cross-dataset generalization. When trained on MIT-BIH and tested on INCART, MHA-MSD achieves 97.38% overall accuracy, with class-wise precision of 96.07% for supraventricular ectopic beats (S) and 99.65% for ventricular ectopic beats (V).

This study aims to develop and evaluate MHA-MSD for ECG classification, with an emphasis on cross-dataset generalization. We consider nine CPSC2018 diagnostic categories: normal ECGs (Normal), atrial fibrillation (AF), first-degree atrioventricular block (I-AVB), left and right bundle branch blocks (LBBB/RBBB), premature atrial and ventricular contractions (PAC/PVC), and ST-segment depression/elevation (STD/STE). We also evaluate normal (N), supraventricular ectopic (S), and ventricular ectopic (V) beats in the MIT-BIH-to-INCART transfer, as well as N, LBBB, and RBBB across CPSC2018, PTB-XL, and CODE-15%.

The paper is organized as follows. Section 1 introduces the research background. Section 2 presents the related work. Section 3 provides a detailed description of the proposed method and experimental setup. Section 4 reports the experimental results. Section 5 discusses the findings, and Section 6 presents the conclusions.

2. Related Work

ECG classification has evolved from handcrafted feature engineering and conventional classifiers [5] to end-to-end deep learning [11,12]. Recent studies have focused on combining complementary representations: Farag [13] integrated time-frequency information using an STFT-based CNN, Alamatsaz et al. [14] combined CNN feature extraction with LSTM temporal modeling, and Kuila et al. [15] employed an LSTM-based architecture for

arrhythmia classification. Chen et al. [16] further demonstrated the value of multi-domain CNN feature fusion for cardiac disease recognition.

Beyond general-purpose architectures, ECG studies have targeted different arrhythmias using task-specific signal representations. For atrial fibrillation (AF), Cai et al. [17] used a densely connected network on raw 12-lead ECGs, whereas Krasteva et al. [18] converted Holter recordings into ECHOView images for transfer learning with pretrained networks. To incorporate explicit rhythm and morphology cues, Khamis et al. [19] combined handcrafted RR-interval and PQRST features with a neural network ensemble. For ventricular arrhythmias, Liu et al. [20] classified ventricular fibrillation (VF) and ventricular tachycardia (VT) from short single-lead ECG segments using a one-dimensional CNN without QRS detection. Focusing on ventricular ectopic beats, Zhou et al. [6] combined convolutional and LSTM networks with rule-based inference to detect premature ventricular contractions (PVCs). For conduction disorders, Ribeiro et al. [12] employed a deep residual network to identify first-degree atrioventricular block, left bundle branch block (LBBB), and right bundle branch block (RBBB) from 12-lead ECGs. To reduce the need for beat-level annotations, Hu et al. [21] adopted multi-instance learning for LBBB and RBBB detection. These studies highlight the diversity of ECG representations and modeling strategies required for different diagnostic tasks.

Beyond these task-specific methods, attention-based architectures have been developed to model long-range relationships in ECG signals. Building on the Transformer architecture [22], Tao et al. [23] developed a refined self-attention Transformer for arrhythmia detection, Kim et al. [24] combined local CNN features with Transformer-based global representations, and Chen et al. [25] incorporated time-frequency characteristics into a Swin Transformer. Related hybrid designs have also fused temporal features or data augmentation with Transformer representations [26,27]. These studies emphasize the importance of global feature modeling, while adaptive integration of features across different temporal scales remains an important consideration.

Meanwhile, self-supervised learning and foundation models have sought more transferable ECG representations. Kiyasseh et al. [28] learned cardiac representations across signals, time, and patients, while Wu et al. [29] used unlabeled signals for self-supervised ECG delineation. More recently, Vaid et al. [30] and McKeen et al. [31] demonstrated the potential of large-scale pretrained models for generalizable ECG interpretation. To address cross-dataset distribution shifts, Li et al. [4] introduced domain-invariant feature enhancement and attention constraints into a Transformer-based ECG classification framework. Despite these advances, robust cross-dataset generalization remains challenging due to differences in signal characteristics and data distributions. Motivated by these considerations, this study combines multihead self-attention, adaptive multiscale dilated convolution, input-dependent gating, and spatial attention fusion to capture long-range dependencies and multiscale ECG features.

3. Materials and Methods

3.1. Problem Formulation

The task involves detecting and classifying arrhythmias using ECG time-series signals. The input dataset is defined as $D = \{(\mathbf{x}_i, y_i) \mid i = 1, 2, \ldots, N\}$. Here, $\mathbf{x}_i$ represents an ECG recording or signal segment rather than a single sampling value. It is represented as a two-dimensional numerical matrix of size $C_{\text{in}} \times L_0$, where C_{in} denotes the number of ECG leads and L_0 denotes the number of temporal sampling points. Each row corresponds to one lead, and each column corresponds to one sampling point. The matrix contains digitized ECG amplitudes rather than image pixels. For a 12-lead input, $C_{\text{in}} = 12$. y_i is its true label, and N is the total number of samples. Labels satisfy $y_i \in \{0, 1, 2, \ldots, C-1\}$, where C is the number of classes.

The value of C depends on the classification task. For CPSC2018, single-label classification is performed across nine classes ($C = 9$): Normal, AF, I-AVB, LBBB, RBBB, PAC, PVC, STD, and STE. For MIT-BIH-to-INCART transfer, three heartbeat classes ($C = 3$) are considered: normal (N), supraventricular ectopic (S), and ventricular ectopic (V) beats. For cross-dataset classification among CPSC2018, PTB-XL, and CODE-15%, three diagnostic classes ($C = 3$) are considered: N, LBBB, and RBBB. The classifier produces C class scores for each task. The loss function uses cross-entropy between true labels and model outputs. Model parameters are optimized to minimize this loss.

The same MHA-MSD architecture was used across all experimental tasks, while the output dimension of the final classifier was configured according to the number of target classes in each task.

3.2. Model Overview

As shown in Figure 1, the proposed MHA-MSD model comprises four components:

1. Front-end Network: This component comprises a CNN-based backbone and an encoder. It performs fundamental feature extraction on ECG signals. Temporal dynamics and inter-lead correlations are explicitly encoded within this stage.
2. Multihead Self-Attention Encoder: Position-encoded features from the front-end network are projected into a high-dimensional space. Multihead self-attention mechanisms vectorize feature relationships. Geometric interactions between embedded vectors are computed as contextual features.
3. Adaptive multiscale Dilated Convolution Module: A gating mechanism dynamically assigns weights to four parallel branches. Each branch operates with distinct dilation rates. This adaptively adjusts the significance of multiscale features before fusion.
4. Spatial Attention Fusion Module: This module fuses features from the weighted branches. It employs spatial attention for feature refinement. Dimensionality reduction occurs along the branch dimension. The condensed feature map is then fed into the classifier.

For model interpretability analysis, one-dimensional gradient-weighted class activation mapping (1D Grad-CAM) was applied to the final convolutional layer of MHA-MSD to generate class-discriminative heatmaps over ECG waveforms.

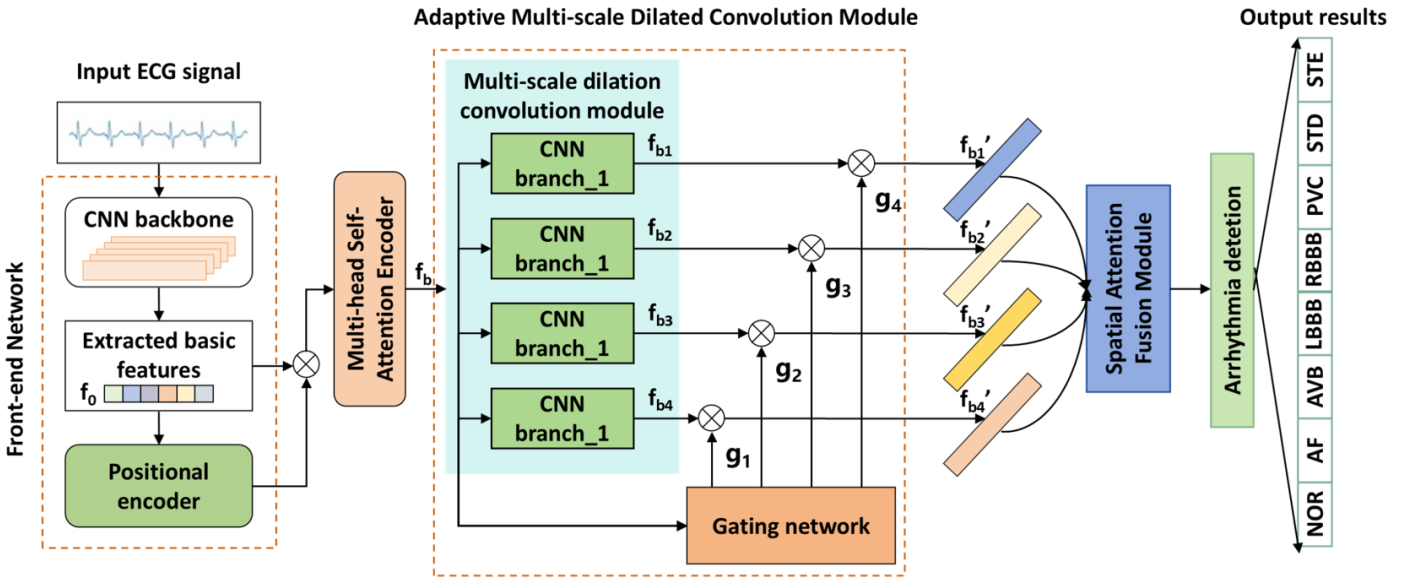


Figure 1. Overview of the model architecture.

3.3. Front-End Network

The front-end network integrates a CNN-based backbone and an encoder. As depicted in Figure 2, the backbone consists of seven convolutional layers. It incorporates batch normalization (BN), activation layers, and pooling operations. Convolutional filters progressively increase in number. This design hierarchically extracts fundamental features from input ECG signals. Max-pooling downsamples feature maps. It retains salient characteristics while reducing dimensionality. BN layers stabilize training by regulating gradient flow. They alleviate the problems of exploding and vanishing gradients in deep networks. ReLU activation introduces nonlinearity. This enhances convergence on nonlinear problems. It also reduces overfitting risks in deep architectures. The backbone processes input $\mathbf{x}_i$ into base feature $\mathbf{f}_b$:

$$\mathbf{f}_b = N_b(\mathbf{x}_i; \theta_b) \tag{1}$$

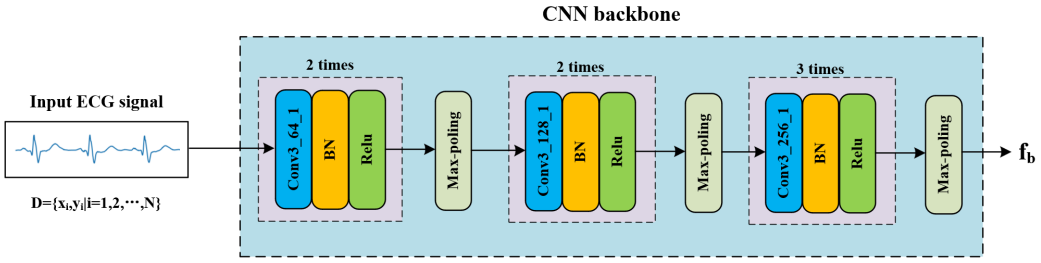


Figure 2. Backbone network structure diagram.

The base feature $\mathbf{f}_0$ output from the backbone network is fed into an encoder. The positional encoder then encodes $\mathbf{f}_0$ along two dimensions: time and ECG leads. This strengthens temporal features within the feature map and establishes connections between features from different leads at the same timestep, facilitating the extraction and learning of inter-lead features in ECG signals.

The input-channel dimension of the CNN backbone is adapted to the ECG lead configuration of each classification task. For the MIT-BIH-to-INCART cross-dataset heartbeat classification, the two recorded channels of MIT-BIH were used, while leads II and V5 were selected from the 12-lead INCART recordings, resulting in an input shape of $[128, 2, L_0]$. For the 12-lead ECG classification experiments, the input shape was $[128, 12, L_0]$. Accordingly, the input can be generally represented as $[128, C_{\text{in}}, L_0]$, where C_{in} denotes the number of input channels and L_0 denotes the temporal length of the ECG signal. The backbone produces an output feature map $\mathbf{f}_0$ of shape $[128, 256, L_1]$, where L_1 denotes the temporal length of the output feature map.

For temporal encoding, the positional encoder processes each of the 128 $[256, L_1]$ feature maps along the time dimension L_1. It applies cosine encoding to the odd columns and sine encoding to the even columns of the $[256, L_1]$ feature maps. This enhances the model's ability to capture multiscale temporal patterns using different frequency components.

For inter-lead encoding, the positional encoder first creates indices for different leads at the same timestep. It then generates cross-lead learnable embedding vectors based on these indices. These embeddings are updated during training to capture combinations of features across leads.

3.4. Multihead Self-Attention Encoder

The multihead self-attention encoder comprises a self-attention layer and a subsequent feedforward network (FFN), with its architecture illustrated in Figure 3. Causal masking is implemented in both layers by setting upper-triangular elements to negative infinity.

This design explicitly constrains the visible regions for the multihead encoder and FFN, enhancing feature causality while filtering invalid timesteps.

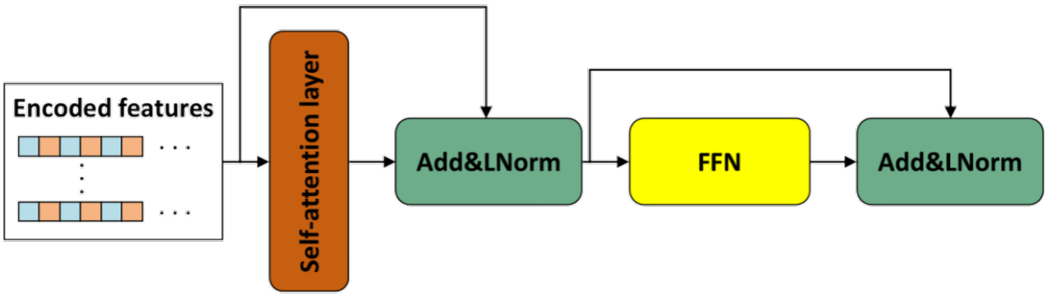


Figure 3. Structure diagram of multihead self-attention encoder.

The structure of the self-attention layer is shown in Figure 4. Within the self-attention layer, encoded input features are processed by an 8-head multihead encoder. These features are projected into a high-dimensional space, where key (**K**), value (**V**), and query (**Q**) vectors are computed for each element. Each attention head independently calculates these vectors and derives the encoder's output features. The scaled dot product between **Q** and **K** vectors is computed per head, followed by softmax normalization to generate attention weights. These weights are then applied to the **V** vectors to produce weighted sums. Outputs from all eight heads are concatenated to form the final representation. The scaled dot-product attention for a single head is defined as follows:

$$\text{Multihead}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{Concat}(\text{head}_1,\ldots,\text{head}_b) \times \mathbf{W}^O \tag{2}$$

$$\text{head}_i = \text{Attention}(\mathbf{Q}\mathbf{W}_i^Q, \mathbf{K}\mathbf{W}_i^K, \mathbf{V}\mathbf{W}_i^V) \tag{3}$$

$$\text{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \tag{4}$$

where $\mathbf{W}^O$ projects concatenated features linearly. Scaling by $\sqrt{d_k}$ prevents gradient vanishing caused by excessive dot product magnitudes in high-dimensional spaces.

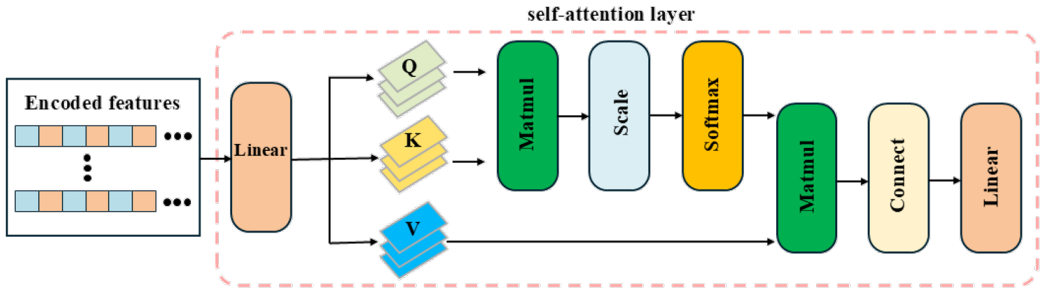


Figure 4. Structure diagram of the self-attention layer.

The FFN is positioned after the self-attention layer within the multihead encoder. Given that ECG signals inherently contain numerous nonlinear features, incorporating the nonlinear ReLU activation function into the linear transformation process of the feedforward network enables the model to effectively capture these nonlinear characteristics while maintaining linear convergence, thereby further enhancing the model's classification performance. The linear combination of the feedforward network's activation functions is detailed in Equation (5), where the trainable parameters W_{ff} and b_{ff} satisfy $W_{ff}^1 \in \mathrm{R}^{d_{\text{model}} \times d_{ff}}$ and $W_{ff}^2 \in \mathrm{R}^{d_{ff} \times d_{\text{model}}}$

$$\text{FFN}(F') = \text{ReLU}(F'W_{ff}^1 + b_{ff}^1)W_{ff}^2 + b_{ff}^2 \tag{5}$$

3.5. Adaptive Multiscale Dilated Convolution Module

The features $\mathbf{f}_b$ output by the feedforward network are simultaneously fed into gating network and four branch networks. As shown in Figure 5, these branch networks share identical architectures and parameters except for their dilation rates (d_j) and kernel sizes (k).

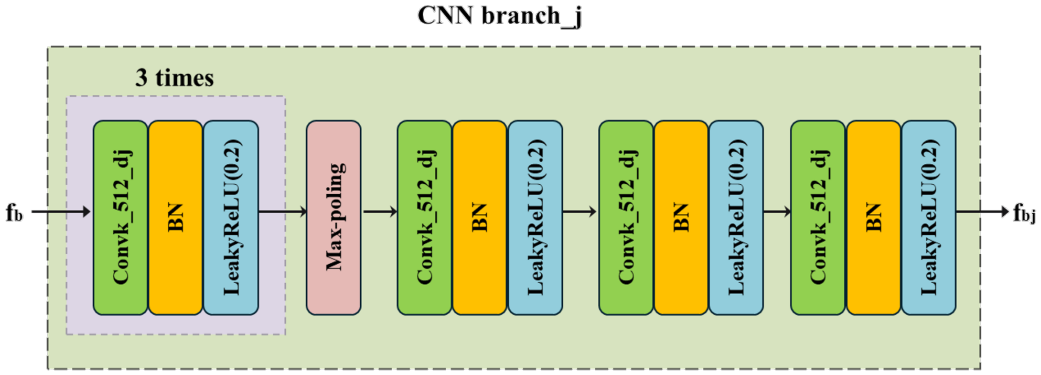


Figure 5. Branch network structure diagram.

The kernel size k is set to 3 for branches $j = 1$, $j = 2$, and $j = 3$ and 4 for branch $j = 4$. For dilation rates, $d_4 = 1$ for the fourth branch, while d_1, d_2, and d_3 are configured as coprime integers according to Equations (6) to (8), where L denotes the current input signal length and $[x]$ represents the floor operation. This configuration implements the Hybrid Dilated Convolution principle [32], assigning signal-length-adapted small (d_1), medium (d_2), and large (d_3) dilation rates to capture multiscale ECG features. The fourth branch ($j = 4$) employs conventional dilation ($d_4 = 1$) with enlarged kernel size to generate complementary receptive fields.

Given the effective receptive field $R = (k-1)d + 1$ of a dilated convolution, the kernel size of 3 in the first three branches enables temporal receptive fields to be adjusted through the input-length-dependent dilation rates without increasing the kernel size. In contrast, the fourth branch employs a kernel size of 4 with unit dilation to retain contiguous local feature extraction, complementing the broader temporal representations of the dilated branches.

Before feature fusion, the gating network dynamically weights and continuously optimizes each branch's contributions. The transformation from base features $\mathbf{f}_b$ to pre-fusion features $\mathbf{f}_{b_j}$ is formalized in Equation (9).

$$d_1 = 1 \tag{6}$$

$$d_2 = [0.5 \times (\frac{L}{1000})] + 1 \tag{7}$$

$$d_3 = [\frac{L}{1000}] \tag{8}$$

$$\mathbf{f}_{b_j} = N_{b_j}(x_i; \theta_{b_j}) \tag{9}$$

The features $\mathbf{f}_b$ output by the feedforward network are simultaneously fed into both the gating network (illustrated in Figure 6) and the four branch networks. The gating network processes $\mathbf{f}_b$ to perform additional feature extraction, and it then outputs a four-dimensional vector through a softmax activation layer. This vector is formed by concatenating four one-dimensional arrays. After dimensionality splitting, the gating network generates four weighting coefficients that multiply the feature vectors output by the four branch networks, respectively. Throughout model training, these weights are continuously updated along the direction of cross-entropy loss reduction and model gradient descent. This optimization ensures progressively increasing representation of effective features in the fused feature map

during feature integration, ultimately achieving the optimal receptive field combination for the given data. Mathematically, the gating network's processing of $\mathbf{f}_b$ is formalized in Equation (10), where $\mathbf{g}$ represents the pre-split four-dimensional vector, $\mathbf{f}_g$ denotes the further-extracted feature vector, and θ_g signifies the gating network's trainable parameters.

$$\mathbf{g} = \text{Softmax}\left(\mathbf{f}_g(\mathbf{f}_b; \theta_g)\right) \tag{10}$$

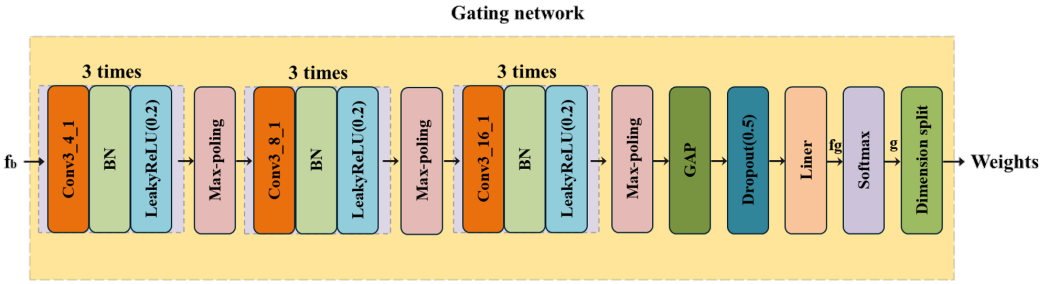


Figure 6. Gating network structure diagram.

3.6. Spatial Attention Fusion Module

Following weighting by the gating network, the branch outputs $\mathbf{f}'_{b_1}$, $\mathbf{f}'_{b_2}$, $\mathbf{f}'_{b_3}$, and $\mathbf{f}'_{b_4}$ enter the feature fusion network and spatial attention module in Figure 7. The Cat operation connects these four feature vectors along the second dimension, generating fused feature map $\mathbf{f}_c$. Within the spatial attention module, $\mathbf{f}_c$ undergoes simultaneous global average pooling and global max pooling. This extracts both its average features and most prominent characteristics. These pooled results are then concatenated via Cat and processed through a 1×1 convolutional kernel. The convolution outputs a spatial attention feature map $\mathbf{f}_{a_0}$. Next, $\mathbf{f}_{a_0}$ and $\mathbf{f}_c$ undergo weighted summation to produce attention-enhanced feature map $\mathbf{f}_a$. Feature map $\mathbf{f}_a$ then passes through average pooling to create smoothed feature $\mathbf{f}_z$. Finally, $\mathbf{f}_z$ feeds into the classifier for prediction. Equation (11) defines the computation of $\mathbf{f}_a$, where $\otimes$ denotes channel-wise multiplication. Equation (12) calculates $\mathbf{f}_z$ using the average pooling function $g_a(x)$. The loss function L in Equation (13) uses total samples N and label categories C. Equation (14) determines probability p, representing the model's likelihood of assigning label y_i to input x_i, with $\mathbf{w}_k$ as class-k parameters.

$$\mathbf{f}_a = \mathbf{f}_c + \mathbf{f}_{a_0} \otimes \mathbf{f}_c \tag{11}$$

$$\mathbf{f}_z = g_m(\mathbf{f}_a) \tag{12}$$

$$L = -\frac{1}{N} \sum_{i=1}^{N} \sum_{k=0}^{C} I\{y_i = k\} \log p(\mathbf{f}_z) \tag{13}$$

$$p = \frac{\exp(\mathbf{w}_{y_i}^T \mathbf{f}_z)}{\sum_{k=1}^{C} \exp(\mathbf{w}_k^T \mathbf{f}_z)} \tag{14}$$

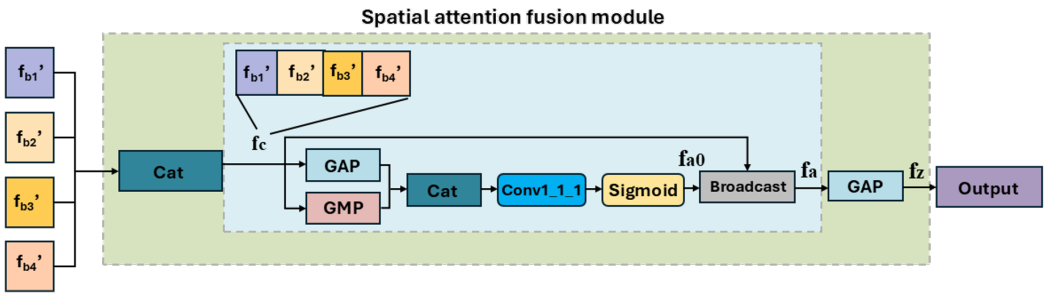


Figure 7. Structure diagram of the attention fusion module.

3.7. Datasets

To comprehensively evaluate our proposed model, we utilized five public ECG datasets for training and validation, including MIT-BIH [7], CPSC2018 [33],INCART [8], PTB-XL [9] (a large electrocardiography dataset), and CODE-15% [10] (a large-scale annotated 12-lead ECG dataset).In the MIT-BIH dataset, supraventricular ectopic (S) and ventricular ectopic (V) beats are less frequent than normal (N) beats.

MIT-BIH contains two-channel ECG recordings sampled at 360 Hz with beat-level annotations; INCART contains 12-lead recordings sampled at 257 Hz with beat-level arrhythmia annotations. CPSC2018 and the selected PTB-XL recordings are 12-lead ECG signals sampled at 500 Hz, whereas CODE-15% provides 12-lead recordings sampled at 400 Hz; these three datasets provide record-level diagnostic labels.

Three experimental settings were considered: nine-class within-dataset classification on CPSC2018, N/S/V heartbeat-level cross-dataset classification between MIT-BIH and INCART, and N/LBBB/RBBB record-level cross-dataset classification among CPSC2018, PTB-XL, and CODE-15%.

For data separation, the CPSC2018 nine-class experiment used 10-fold cross-validation; MIT-BIH was partitioned at the patient level into training, validation, and test sets at a 7:2:1 ratio, with INCART used as the external test set.

For the record-level cross-dataset experiments, CPSC2018 was randomly stratified into training, validation, and test sets at a 7:2:1 ratio. PTB-XL followed its recommended patient-level stratified partition, with folds 1–8 used for training, fold 9 for validation, and fold 10 for testing [9]. For CODE-15% [10], the selected records were partitioned into training, validation, and test sets at an 8:1:1 ratio.

Each dataset was used as the source and target in turn, yielding six cross-dataset transfer configurations. For each configuration, the model was trained on the source training set, validated on the source validation set, and evaluated on the target test set.

3.8. Evaluation Metrics

To evaluate classification performance, we employ accuracy, precision, recall, and F1 score. For each class k, true positives (TP_k), false positives (FP_k), and false negatives (FN_k) are defined using a one-versus-rest approach. These metrics are calculated as follows:

$$\text{Accuracy} = \frac{N_{\text{correct}}}{N_{\text{total}}} \tag{15}$$

$$\text{Precision}_k = \frac{TP_k}{TP_k + FP_k} \tag{16}$$

$$\text{Recall}_k = \frac{TP_k}{TP_k + FN_k} \tag{17}$$

$$F1_k = \frac{2\,\text{Precision}_k\,\text{Recall}_k}{\text{Precision}_k + \text{Recall}_k} \tag{18}$$

$$\text{Mean F1} = \frac{1}{C}\sum_{k=0}^{C-1} F1_k \tag{19}$$

Here, N_{correct} and N_{total} denote the numbers of correctly classified samples and all evaluated samples, respectively. For each class k, TP_k represents correctly identified samples, FP_k represents samples incorrectly assigned to class k, and FN_k represents samples of class k incorrectly assigned to other classes. Overall accuracy measures classification performance across all evaluated ECG samples, whereas precision, recall, and F1 score are calculated separately for each class. Mean F1 is the macro-average of class-wise F1 scores, giving equal weight to each class.

3.9. Training Details

3.9.1. Training Environment

The model was implemented in PyTorch (version 2.5.1) and trained on an NVIDIA RTX 4090D GPU with a batch size of 128. Stochastic gradient descent (SGD) with momentum 0.9 and an initial learning rate of 0.001 was used for optimization. Kaiming initialization was applied, and the checkpoint with the highest validation mean F1 score was retained.

3.9.2. Preprocessing and Data Augmentation

Baseline drift was removed using a fourth-order Butterworth high-pass filter with a 0.75-Hz cutoff, and high-frequency noise was suppressed through four-level db6 stationary wavelet decomposition and reconstruction. For the MIT-BIH-to-INCART heartbeat-level classification experiment, the reference beat annotations provided by both databases were used as alignment anchors. MIT-BIH recordings were processed at their native sampling rate of 360 Hz, while INCART recordings were resampled from 257 Hz to 360 Hz before heartbeat extraction. The annotation positions were adjusted to the resampled time axis. In both datasets, each selected annotation position x was used as the anchor, and a 301-sample segment spanning $x - 100$ to $x + 200$ (inclusive) was extracted. At 360 Hz, this corresponds to approximately 0.278 s before and 0.556 s after the anchor. For consecutive annotated beats with an RR interval of 200 samples or less, an overlap-filtering procedure was applied to the extracted heartbeat segments. The segment associated with the beat exhibiting the higher QRS amplitude was retained, while the other candidate segment was discarded. Class imbalance was addressed by undersampling N-class beats and applying diffusion-based augmentation to minority classes. In the MIT-BIH training set, 4000 synthetic S-class samples increased this class from 1945 to 5945 samples. Figure 8 illustrates preprocessing and augmentation examples.

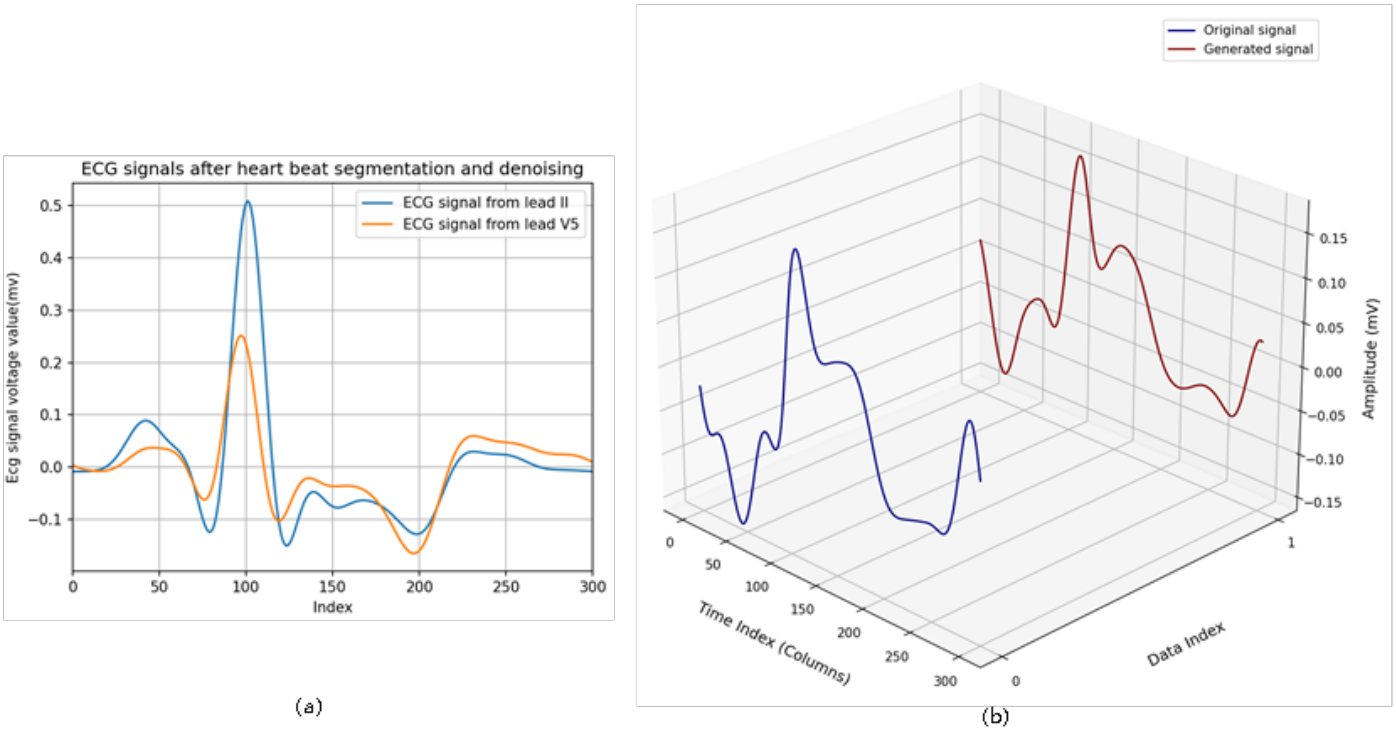


Figure 8. Examples of ECG preprocessing and data augmentation. (**a**) Filtering and beat segmentation of the first effective beat from MIT-BIH record No. 100, showing the MLII and V5 channels. (**b**) Data0 is an original S-class beat from the MLII (modified lead II) channel, and Data1 is a diffusion-generated S-class beat corresponding to the same channel.

The preprocessing procedures were configured according to the experimental task. CPSC2018 recordings were used at 500 Hz for nine-class classification. For the cross-dataset experiments among CPSC2018, PTB-XL, and CODE-15%, CPSC2018 and the selected PTB-XL recordings were used at 500 Hz, while CODE-15% recordings were resampled from 400 Hz to 500 Hz; records labeled N, LBBB, and RBBB were retained. For the MIT-BIH-to-INCART experiment, N, S, and V heartbeat annotations were selected, with leads II and V5 used from the INCART recordings. The recording-level experiments used full ECG recordings without R-peak-centered heartbeat segmentation.

In Figure 8b, Data0 and Data1 both correspond to the MLII (modified lead II) channel; they are not signals from two different leads. Data0 is an original S-class beat, whereas Data1 is a diffusion-generated S-class beat for the same channel. S-class beats may have morphology that differs from the representative beat shown in Figure 8a.

4. Results

4.1. Accuracy Validation on CPSC2018

To address the non-public availability of the CPSC2018 test set, we implemented 10-fold cross-validation by randomly dividing the public training data into 10 approximately equal folds. For each iteration, one distinct fold served as the test set while the remaining nine formed the training set. Seven iterations used 688 test records and 6,189 training records each, while the other three used 687 test records and 6,190 training records each. Each record was used for testing once across the 10 folds. The results of the fold achieving the highest macro-F1 score among the ten cross-validation iterations are presented in Table 1, with the corresponding confusion matrix shown in Figure 9.

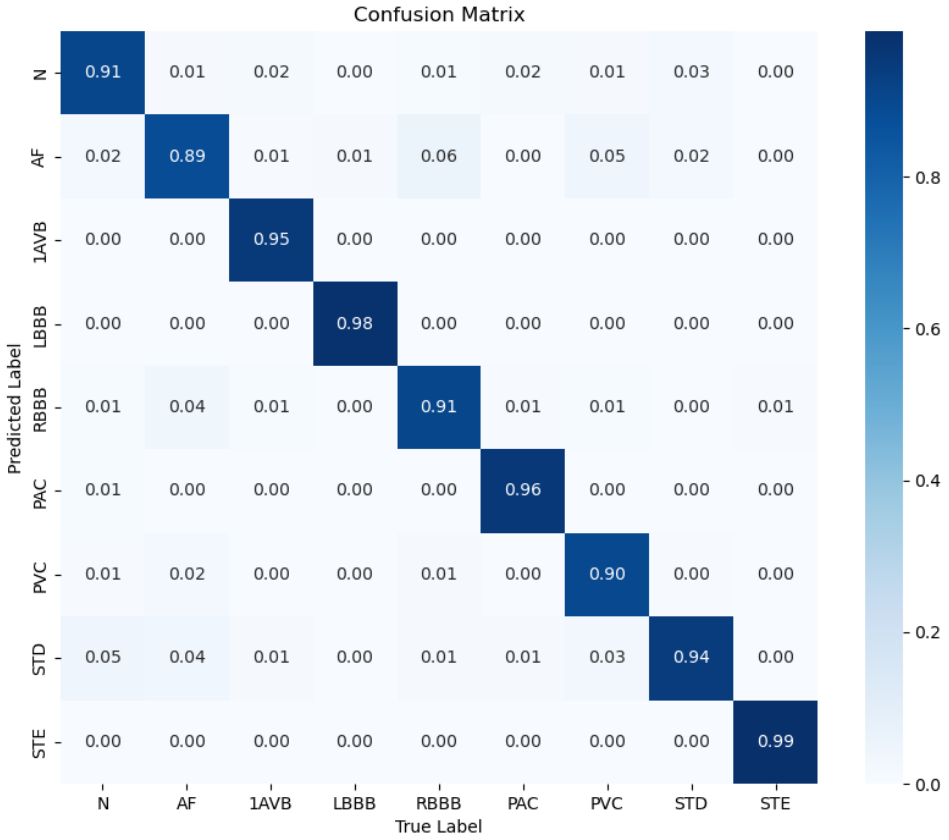


Figure 9. Confusion matrix for the best-performing fold of 10-fold cross-validation on CPSC2018.

Table 1. Best-fold results of 10-fold cross-validation on the CPSC2018 dataset.

Class	Precision	Recall	F1 Score
Normal	0.909	0.868	0.888
AF	0.886	0.826	0.855
I-AVB	0.950	1.000	0.975
LBBB	0.981	0.990	0.986
RBBB	0.909	0.945	0.926
PAC	0.959	0.993	0.976
PVC	0.902	0.904	0.903
STD	0.944	0.789	0.860
STE	0.992	0.998	0.996
Accuracy			0.936
Mean F1 Score			0.929

Note: "Mean F1 Score" denotes the macro-average of the nine class-wise F1 scores for the selected folds.

Experimental results demonstrate that our proposed model achieves precise detection and classification of multilabel ECG signals through effective feature extraction. Compared to existing models, it shows superior performance, evidenced by higher F1 scores across all categories in multiclass classification, as detailed in Table 2. Against Ahmed Mostayed's Bi-directional LSTM Network [34], our model exhibits significant improvements for all nine diagnostic categories, with F1 scores increasing by 9.7% for AF and 9.6% for PVC labels and over 10% for other labels. Against Yongbo Liang's CNN and BiLSTM [35], substantial gains are observed in most categories, particularly a breakthrough 70%+ F1 score improvement for STE detection. Compared to Tsai-Min Chen's CNN-Based Model [36], our approach demonstrates enhanced classification capability across all labels, delivering 48.7% and 16.5% F1 score gains for STE and LBBB categories, respectively, along with varied improvements for other non-AF classes.

Table 2. Comparison of reported class-wise F1 scores on the CPSC2018 dataset.

	F1 Score			
Class	**Bi-Directional LSTM Network [34]**	**CNN and BiLSTM [35]**	**CNN-Based Model [36]**	**MHA-MSD (Ours)**
Normal	0.739	0.762	0.795	0.888
AF	0.768	0.895	0.897	0.855
I-AVB	0.742	0.872	0.865	0.975
LBBB	0.706	0.833	0.821	0.986
RBBB	0.822	0.943	0.911	0.926
PAC	0.591	0.784	0.734	0.976
PVC	0.807	0.821	0.852	0.903
STD	0.658	0.818	0.788	0.860
STE	0.294	0.471	0.509	0.996

Note: The results of the comparison methods were reproduced from their original publications [34–36]. Differences in data partitioning, validation protocols, and preprocessing procedures may exist among these studies. Therefore, the comparison is indicative rather than strictly controlled. The MHA-MSD results correspond to the best-performing fold of 10-fold cross-validation.

To validate the contribution of individual modules, we conducted ablation studies on the CPSC2018 dataset. As shown in Table 3, increasing the number of branches progressively improved performance, while the gating network (Gate), spatial attention fusion (SAF), and multihead self-attention provided further gains. The complete MHA-MSD model achieved the best accuracy and F1 score.

Table 3. Ablation experiments of different modules on CPSC2018.

Model	Accuracy	F1 Score
1Branch	0.497	0.460
2Branchs	0.527	0.479
3Branchs	0.674	0.625
4Branchs	0.793	0.755
4Branchs + Gate	0.829	0.843
4Branchs + Gate + SAF	0.904	0.907
MHA-MSD (ours)	0.936	0.929

4.2. Cross-Dataset Generalization on MIT-BIH and INCART

To evaluate cross-dataset generalization, we used three heartbeat classes: normal beats (N), supraventricular ectopic beats (S), and ventricular ectopic beats (V). After label aggregation, the MIT-BIH dataset contains 90,372 N beats, 2223 S beats, and 5788 V beats, totaling 98,383 heartbeats. The data were split at the patient level into training, validation, and internal test sets at a 7:2:1 ratio. The entire INCART database, comprising 75 annotated 30 min ECG recordings from 32 Holter records, was used as the external test set. Models were trained on its training set, selected using its validation set, and finally evaluated on the entire INCART dataset. Class-wise metrics are reported in Table 4, and the ROC curves are shown in Figure 10.

Table 4. INCART results after training on MIT-BIH.

Class	Precision	Recall	F1 Score
N	0.9613	0.9925	0.9766
S	0.9607	0.9767	0.9686
V	0.9965	0.9583	0.9771
Accuracy		0.9738	
Mean F1 score		0.9741	

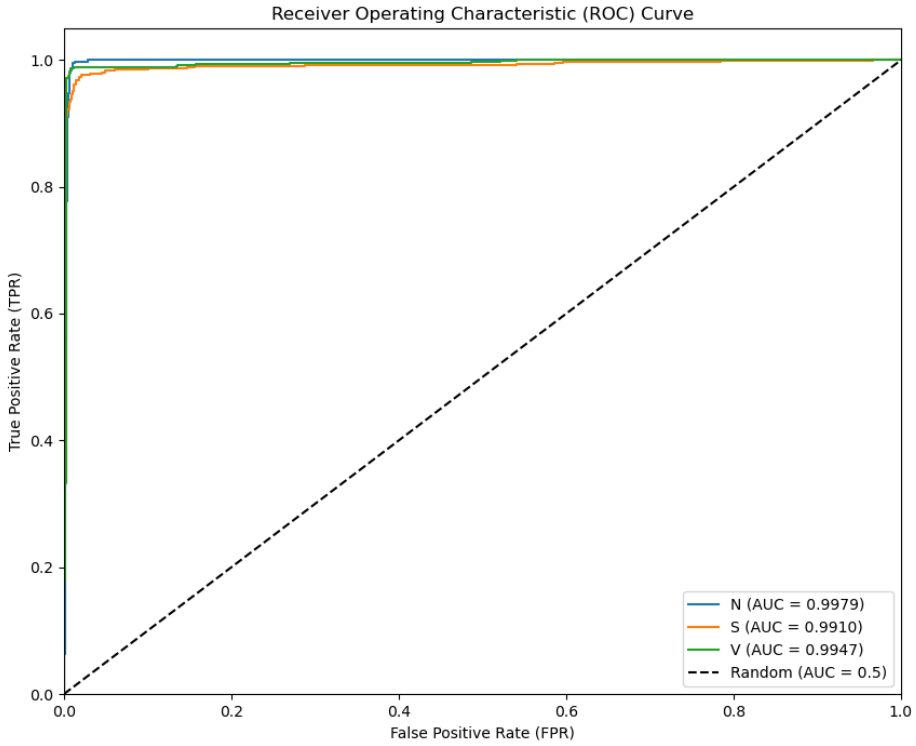


Figure 10. The ROC curve of the model tested on the INCART dataset after training on the MIT-BIH dataset.

To assess generalization capabilities against recent models, we compared our performance with other models trained and tested on identical datasets, as shown in Table 5. Results indicate significant improvements for S-class and V-class classification. Specifically for S-class, our proposed MHA-MSD model achieved the highest performance among four benchmark models, surpassing the second-ranked ERCA model [37] by 18.37%. For V-class classification, our model showed a 1.31% improvement over Wang et al.'s DAEAC model [3] and greater gains against others. We observed a marginal decrease in N-class classification and overall accuracy compared to other models. This slight degradation may stem from diffusion-based augmentation for S-class data: some generated samples exhibited blurred features or feature blending, increasing misclassification between N-class and S-class. Since N-class dominates the INCART dataset, overall accuracy was consequently impacted.

Table 5. Comparison of the test results of different models trained on the MIT-BIH dataset and then tested on the INCART dataset.

Class	Precision			Overall Accuracy
	Normal Beats (N)	Supraventricular Ectopic Beats (S)	Ventricular Ectopic Beats (V)	
DDCNN [38]	0.9960	0.1440	0.8190	0.9160
DAEAC [3]	0.9335	0.4057	0.9834	0.9270
ERCA [37]	0.9886	0.7770	0.9392	0.9813
MHA-MSD (ours)	0.9613	0.9607	0.9965	0.9738

4.3. Cross-Dataset Generalization on CPSC2018, PTB-XL, and CODE-15%

To comprehensively validate classification and generalization capabilities across multiple datasets, we conducted cross-validation on three public resources: CPSC2018, PTB-XL, and CODE-15%. All datasets were uniformly resampled to 500 Hz. We extracted records originally labeled N, LBBB, and RBBB from each dataset. For data partitioning, PTB-XL and CODE-15% followed their official stratified sampling protocols for train/validation/test sets. CPSC2018, lacking official guidelines, underwent random stratified partitioning using a 7:2:1 ratio. For the cross-dataset experiments, the training sets contained 4814, 17,470, and 1863 records from CPSC2018, PTB-XL, and CODE-15%, respectively. The corresponding test sets contained 1375, 2183, and 233 records, respectively. The sample sizes for each cross-dataset transfer experiment are presented in Table 6. Cross-validation results appear in Table 6.

Table 6. The results of cross-dataset cross-validation of the model on the CPSC2018, PTB-XL, and CODE-15% datasets.

			F1 Score		
Data	**Training Records**	**Test Records**	**N**	**LBBB**	**RBBB**
Train-CPSC2018; Test-PTB-XL	4814	2183	0.87	0.96	0.86
Train-CPSC2018; Test-CODE-15%	4814	233	0.77	0.89	0.79
Train-PTB-XL; Test-CPSC2018	17,470	1375	0.80	0.96	0.86
Train-PTB-XL; Test-CODE-15%	17,470	233	0.70	0.77	0.78
Train-CODE-15%; Test-CPSC2018	1863	1375	0.79	0.99	0.81
Train-CODE-15%; Test-PTB-XL	1863	2183	0.80	0.96	0.72

Table 6 demonstrates our model's sustained generalization capability during multi-dataset cross-validation. Performance peaked in CPSC2018-to-PTB-XL validation, achieving 0.96 and 0.86 F1 scores for LBBB and RBBB classes, respectively. However, RBBB classification notably declined during PTB-XL-to-CODE-15% validation. This performance

gap stems from inherent dataset discrepancies: both PTB-XL and CODE-15% employ label hyperclustering that subsumes LBBB/RBBB categories under broader annotations, resulting in insufficient representative signals. Combined with significant inter-dataset feature variations, these constraints limited classification efficacy for specific cardiac conditions.

4.4. Model Interpretability Analysis

The Grad-CAM visualizations reveal different activation patterns across the N, S, and V heartbeat classes.

For normal beats (N), as shown in Figure 11a, the strongest activation localizes to the P-wave interval, with weaker activation over the QRS complex, consistent with sinus-rhythm reading, in which P-wave integrity is the primary cue.

For supraventricular ectopy (S), as shown in Figure 11b, activation spreads across the P-wave and the QRS onset, reflecting the short RP interval and retrograde P-wave features used to identify supraventricular ectopy on lead II.

For ventricular ectopy (V), as shown in Figure 11c, the heatmap concentrates on the widened QRS, with secondary activation over the ST segment and T-wave, matching the broad QRS and ST/T abnormalities of ventricular origin.

Across the three classes, the attended regions coincide with the segments a clinician examines first, indicating that the representation, including the global context introduced by the self-attention encoder, reflects clinically grounded rather than source-specific features.

Its generalization capability compares favorably with current state-of-the-art models, and the resulting framework offers a generalizable data-driven computational model for computer-aided cardiac decision support.

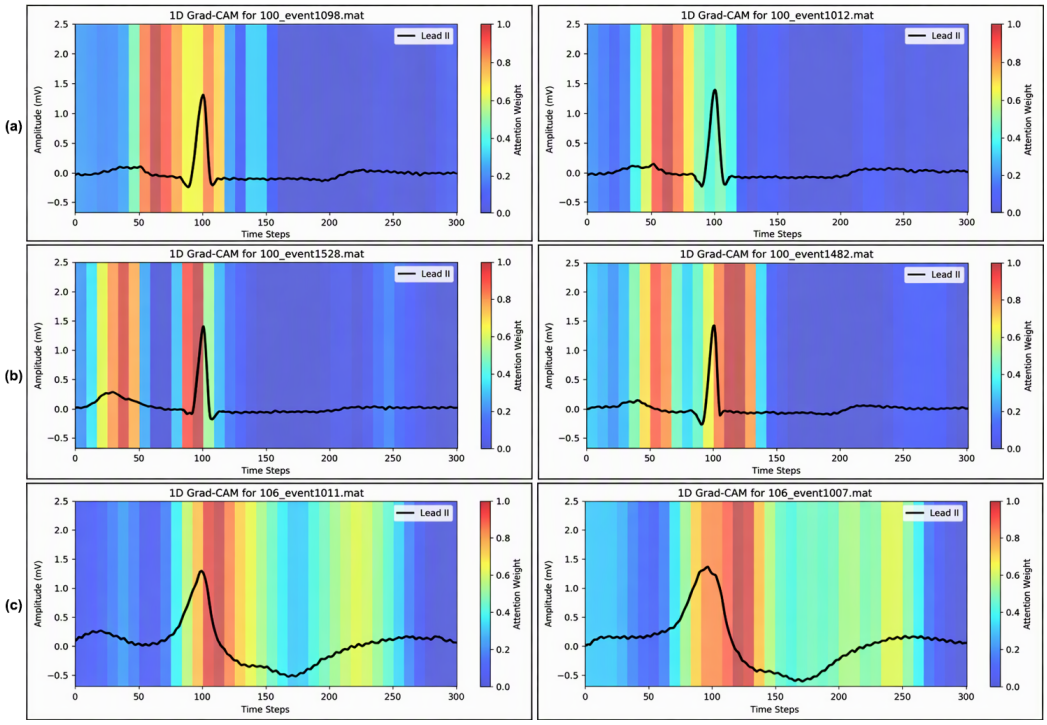


Figure 11. The visualization of the model's attention for classes N, S, and V on the MIT-BIH dataset. (**a**) Class N. (**b**) Class S. (**c**) Class V.

5. Discussion

The cross-dataset experiments indicate that MHA-MSD can transfer across ECG datasets acquired under different conditions, although classification performance varies across source–target pairs. The datasets differ in sampling frequency (257–500 Hz) and lead

configuration (two channels for MIT-BIH versus 12 leads for the other datasets). Recording quality may also be affected by gain, filtering, electrode contact, and motion artifacts. The high-pass filtering and wavelet denoising used in this study were designed to reduce baseline drift and high-frequency noise, but they may not eliminate all acquisition-related variability. Moreover, differences in patient populations and label distributions mean that the current cross-dataset experiments cannot isolate the effects of individual devices or environmental factors. Future work will include controlled tests of acquisition conditions and validation using multi-center, multi-device ECG recordings.

The Grad-CAM visualizations show a qualitative correspondence between the regions emphasized by the model and the P-wave and QRS characteristics relevant to the classification of normal, supraventricular ectopic, and ventricular ectopic beats. These findings provide preliminary evidence of model interpretability, although prospective clinical validation is still required.

Although dilated convolutions enlarge the receptive field without increasing kernel size, the four parallel convolutional branches and multihead self-attention may incur additional computational and memory costs, particularly as ECG sequence length increases. Inference latency, memory consumption, computational requirements, and scalability with input length have not yet been systematically evaluated. Therefore, the suitability of MHA-MSD for real-time clinical deployment remains uncertain. Future work will assess these aspects on resource-constrained devices, quantify fold-to-fold variability in the CPSC2018 cross-validation experiment, and incorporate confidence intervals and statistical significance analyses to better characterize the reported performance.

6. Conclusions

This study presented MHA-MSD, which combines multihead self-attention with gated adaptive multiscale dilated convolution for arrhythmia classification. For CPSC2018, MHA-MSD achieved 93.6% accuracy and a 92.9% mean F1 score in the best-performing fold among the 10 cross-validation iterations, while cross-dataset transfer from MIT-BIH to INCART achieved 97.38% accuracy and a 97.41% mean F1 score. Further cross-dataset experiments among CPSC2018, PTB-XL, and CODE-15% demonstrated F1 scores of 0.87, 0.96, and 0.86 for N, LBBB, and RBBB, respectively, when training on CPSC2018 and testing on PTB-XL. Grad-CAM analysis showed that the model attended to clinically relevant ECG regions. Future work will focus on data-driven receptive-field optimization and quality-control strategies for diffusion-generated samples. Meanwhile, cross-dataset transfer from MIT-BIH to INCART achieved 97.38% accuracy and a 97.41% mean F1 score. Further cross-dataset experiments among CPSC2018, PTB-XL, and CODE-15% demonstrated F1 scores of 0.87, 0.96, and 0.86 for N, LBBB, and RBBB, respectively, when training on CPSC2018 and testing on PTB-XL. Grad-CAM analysis showed that the model attended to clinically relevant ECG regions. Future work will focus on data-driven receptive-field optimization and quality-control strategies for diffusion-generated samples.

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Data Availability Statement: The ECG data used in this study are all publicly available and can be accessed via the following links: MIT-BIH Arrhythmia Database (PhysioNet, https://physionet.org/content/mitdb/1.0.0/accessed on 19 October 2026), China Physiological Signal Challenge 2018 (CPSC2018, PhysioNet, https://physionet.org/content/challenge-2018/1.0.0/accessed on 19 October 2026), St. Petersburg INCART 12-lead Arrhythmia Database (PhysioNet, https://physionet.org/content/incartdb/1.0.0/accessed on 19 October 2026), PTB-XL (PhysioNet, https://physionet.org/content/ptb-xl/1.0.3/accessed on 19 October 2026), and CODE-15 (Zenodo, https://zenodo.org/records/4916206accessed on 19 October 2026).

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